

Title of research article

Analysis and Prediction of the Survival of Titanic Passengers Using Machine Learning

(Book p. 297 to 304): https://link.springer.com/chapter/10.1007/978-981-15-4218-3_29

GitHub Link

<https://github.com/SamuelSiuQUT/MXN442-N11423579-Siu-Wing-Hei>

Research objective

The research article wants to use Titanic data set from Kaggle (<https://www.kaggle.com/competitions/titanic/data>) to predict Titanic passenger survived (1) or not survived (0) . The study will select appropriate features and apply 6 algorithms, logistic regression, KNN, SVM, NaiveBayes, decision tree and random forest. These algorithms will be evaluated both with and without K-Fold cross-validation(k=4,5,10). Finally each model will provide performance metrics by confusion matrix.

Solution technique

Logistic regression: Is good from binary data, use to classify the target variable survive, value is 1 = passenger survived, value is 0 = passenger do not survived.

K-Nearest Neighbors (KNN): KNN classifies Titanic passenger based on the majority class of its k-nearest neighbors in the feature space. Using selected features to calculate the Euclidean distance between new passenger and another passenger in the train set. In the article k = 5, if 3 or more of the nearest neighbors survived, the new passenger is classified as 'survived'."

Support Vector Machine SVM: Use maximizing the margin the margin between classes to separate Titanic passengers into two classes: survived or not, through an optimal hyperplane.

Naïve Bayes: NB assumes that features are independent of each other. The algorithm examines these features one by one to find out how likely they are to belong to a group. After that, the algorithm multiplies these chances together to determine which group has the highest probability, and finally assigns the instance to that group.

Decision Tree: In the Titanic dataset, a decision tree splits the data into smaller parts by asking questions like whether a passenger survived or not. In each split, the decision

tree selects the feature that best separates the data into groups. The algorithm continues splitting until it can easily predict which group the instance belongs to.

Random Forest: A decision tree is just a single tree, while a random forest is made up of multiple decision trees. The algorithm makes predictions by combining the decisions of all the trees. It averages the predictions from each tree, which increases accuracy and helps avoid mistakes made by any one tree.

Article main findings and results

In this article, the authors identify the most important features for predicting Titanic survival, including P-class, sex, age, SibSp, ParCh, and nickname (e.g., titles like Mr., Ms., Master). (SibSp, ParCh “relevant for the group of family units together”). They applied six machine learning algorithms: Logistic Regression, KNN, SVM, Naive Bayes, Decision Tree, and Random Forest, both with and without k-fold cross-validation. For evaluation, the authors used the confusion matrix along with metrics like accuracy, precision, recall, and F1-score to compare the models. Before applying k-fold, Naive Bayes was the best-performing classifier with an accuracy of 98.8%. With k-fold cross-validation, k = 10 outperformed other k fold cross validation models across all classifiers.

Here is the result summary from the article.

Without k-fold cross - validation	Accuracy	Precision	Recall	F1-Score
Logistic Regression	95.45	93.46	94.07	95.45
KNN	86.36	81.45	80.92	86.35
SVM	95.45	92.35	95.39	95.46
Naive Bayes	98.80	99.32	97.36	98.80
Decision Tree	80.62	72.32	75.65	80.71
Random Forest	81.33	72.28	78.94	81.49
K=10	Accuracy	Precision	Recall	F1-Score
Logistic Regression	87.56	83.23	82.64	87.53
KNN	87.27	82.66	82.59	87.24
SVM	86.94	82.59	81.62	86.90
Naive Bayes	86.19	81.86	79.59	86.11
Decision Tree	87.06	82.80	81.35	87.01
Random Forest	86.45	82.21	80.52	86.38
K=5	Accuracy	Precision	Recall	F1-Score
Logistic Regression	85.79	82.76	78.58	85.66
KNN	84.57	80.58	77.84	84.49
SVM	86.25	83.26	79.42	86.12
Naive Bayes	85.71	82.88	78.48	85.51
Decision Tree	81.97	77.68	73.10	81.82
Random Forest	83.27	79.31	75.49	83.16
K=4	Accuracy	Precision	Recall	F1-Score
Logistic Regression	85.48	82.84	78.37	85.38
KNN	83.12	79.18	75.33	83.02
SVM	86.25	83.96	78.97	86.12
Naive Bayes	85.56	83.20	78.23	85.45
Decision Tree	80.59	76.27	71.12	80.42
Random Forest	82.66	78.02	76.03	82.59

Replication of the study

Model building:

For all article six methods and 2 new methods, I used the features Survived ~ Pclass + Sex + Age + SibSp + Parch + Nickname to train the model on the training set, both with and without cross-validation. After training, I used the model to predict outcomes on the test set and validated the results on the validation set using a confusion matrix. Finally, I reported the performance metrics.

Hyperparameters:

Model	Logistic Regression	Logistic Regression 10 K fold cross validation	KNN	KNN 10 K fold cross validation	SVM	SVM 10 K fold cross validation
Hyperparameters	family = binomial	method = "cv", number = 10	k = 5	kmax = 5; distance = 2 kernel = "optimal"	kernel = "radial"	method = "cv", number = 10 method = "svmRadial" tuneLength = 10
Model	Naive Bayes	Naive Bayes 10 K fold cross validation	Decision Tree	Decision Tree 10 K fold cross validation	Random Forest	Random Forest 10 K fold cross validation
Hyperparameters	naiveBayes	method = "nb" method = "cv", number = 10	maxdepth c(5, 10, 20, 25, 30) Try the best maxdepth	method = "cv", number = 10 method = "rpart2" maxdepth c(5, 10, 20, 25, 30) Try the best number	Number of trees: c(10, 100, 250, 500, 1000) Maximum Nodes: c(10, 20, 30, 50, 100) Try the best number	method = "rf" method = "cv", number = 10 Number of trees: c(10, 100, 250, 500, 1000) Maximum Nodes: c(10, 20, 30, 50, 100) Try the best number

Replication Result:

The result of predict Titanic passengers survival rate using different classifiers				
without cross validation				
	Accuracy %	Precision %	Recall %	F1-Score %
Logistic Regression	92.58	91.16	88.16	89.63
K nearest neighbors(KNN)	78.71	69.81	73.03	71.38
Support Vector Machine(SVM) - radial	94.74	90.12	96.05	92.99
Naive Bayes(NB)	97.37	96.08	96.71	96.39
Decision Tree (DT)	93.54	87.43	96.05	91.54
Random Forest(RF)	96.17	95.95	93.42	94.67
The result of predict Titanic passengers survival rate using different classifiers				
with K fold cross validation k = 10				
	Accuracy %	Precision %	Recall %	F1-Score %
Logistic Regression	94.02	88.48	96.05	92.11
K nearest neighbors(KNN)	79.67	71.90	72.37	72.13
Support Vector Machine(SVM) - radial	93.06	88.20	93.42	90.73
Naive Bayes(NB)	66.03	1	6.58	12.35
Decision Tree (DT)	94.74	90.12	96.05	92.99
Random Forest(RF)	97.61	95.51	98.03	96.75
The result of predict Titanic passengers survival rate using different classifiers				
with K fold cross validation k = 5				
	Accuracy %	Precision %	Recall %	F1-Score %
Logistic Regression	94.02	88.48	96.05	92.11
K nearest neighbors(KNN)	79.67	71.90	72.37	72.13
Support Vector Machine(SVM) - radial	93.06	88.20	93.42	90.73
Naive Bayes(NB)	66.03	1	6.58	12.35
Decision Tree (DT)	94.74	90.12	96.05	92.99
Random Forest(RF)	96.65	93.67	97.37	95.48
The result of predict Titanic passengers survival rate using different classifiers				
with K fold cross validation k = 4				
	Accuracy %	Precision %	Recall %	F1-Score %
Logistic Regression	94.02	88.48	96.05	92.11
K nearest neighbors(KNN)	79.67	71.90	72.37	72.13
Support Vector Machine(SVM) - radial	94.02	88.96	95.39	92.06
Naive Bayes(NB)	66.03	1	6.58	12.35
Decision Tree (DT)	94.74	90.12	96.05	92.99
Random Forest(RF)	96.65	93.67	97.37	95.48

Overall difference:

In the paper, all performance metrics result is different. The only metrics that are the same are the SVM's F1-Score (86.12) and Accuracy (86.25) for both $k = 4$ and $k = 5$.

In my results, the performance metrics for models without cross-validation and with cross-validation are different (same with article). However, except for slight variations in SVM and Random Forest, all classifiers produced the same results across different values of k -fold cross-validation.

Difference in detail:

Logistic Regression: In logistic $k=4$ or 5 or 10 have the same performance. With cross validation (CV) performance is better than without CV, except precision. The outcome is different to the article, the article shows the best performance is without CV.

KNN: In KNN, 3 type of k fold CV have the same result. Except the slightly different in Recall. The performance metrics of CV classifiers are better than those of non-CV classifiers. It is the same with the paper.

SVM: In my results, the best performance was achieved without cross-validation, followed by $k = 4$, and finally $k = 10$ and $k = 5$ (which produced the same output). This trend is consistent across all performance. Compared to the article, my SVM with cross-validation consistently performed better across all metrics. In the article, the best performance was also seen without cross-validation, followed by $k = 10$, and then $k = 4$ and $k = 5$.

Naïve Bayes (NB): My best performance with NB is without cross-validation (CV), which is the same as in the article. When cross-validation is applied, the performance metrics drop significantly, and this trend is also observed in the article.

Decision Tree (DT): The performance metrics of my Decision Tree model are better than those in the article. My three types of k -fold cross-validation have the same outcome, and they all perform better than the non-CV method. In the article, the best performance is with $k = 10$.

Random Forest (RF): RF is related to DT, and my RF performance is better than the article. The best method in RF is $k = 10$, followed by $k = 5$ and $k = 4$ (they have same output), lastly is without CV (slightly different with paper). In paper, the best is $k = 10$, then $k = 5$, after than is $k = 4$, lastly is without CV.

Conclusion:

The writers emphasize that Naive Bayes is the top classifier for the Titanic dataset when cross-validation is not applied. Additionally, $k = 10$ performed better than other k -fold cross-validation values. However, the paper does not clearly conclude whether cross-

validation or no cross-validation is better overall, also does not definitively that which classifier is the best across all situations.

In my results, when cross-validation is not applied, Naive Bayes is the best-performing classifier, consistent with the article. Furthermore, I can clearly determine that, except Naive Bayes, $k = 10$ is the best model and the best classifier is Random Forest.

The reason of my performance metrics is similar across different values of k -fold cross-validation, while the papers are not, could be because we used different methods to build the models. Another reason might be that I used `set.seed(1)`, while the paper used a random seed (their code wasn't released). Also, Random Forest performs best in my results because I experimented with different numbers of trees and nodes to optimize the model, which the paper may not have done.

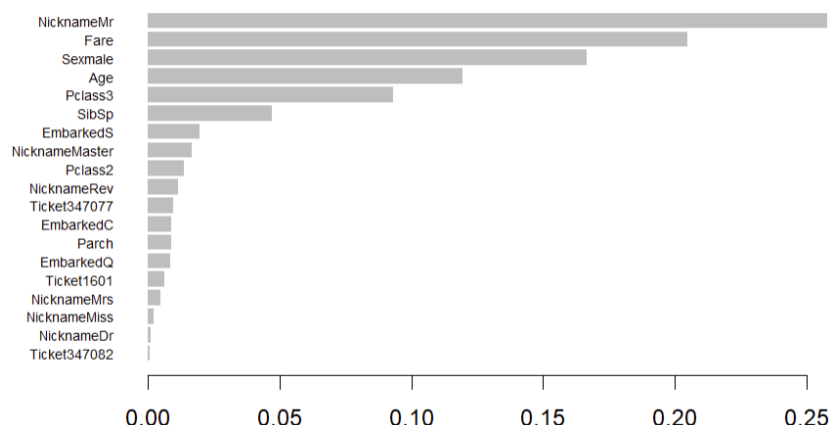
Add two new methods with building new features

I have added two new methods which is Gradient Boosting Machine-Extreme Gradient Boosting (GBM-XGBoost) and Neural Networks. Include without cross validation and 10 k fold cross validation (only $k = 10$, since it performs the best)

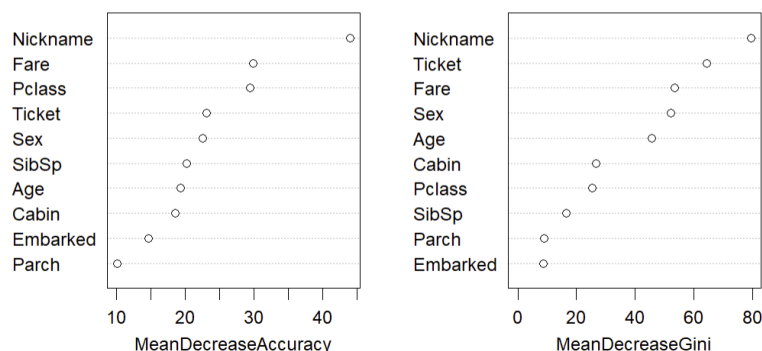
Compare to article 6 method. XGBoost can handles missing values, and reduce overfitting through regularization, so it is more robust for complex data. Neural Networks is more flexible, can capture nonlinear relationships and customizable architectures. These two models enhance classification performance.

Select new features:

In XGBoost I use feature importance-based selection to select new features, which is Survived ~ Pclass + Sex + Age + SibSp + Fare + Nickname (different to article features).



In Neural Networks, first Random Forest do well with binary issue because it builds decision trees that split based on features values. So I decide to use Random Forest to do variable selection.



In the Gini diagram, the features Nickname, Fare, Sex, and Age show strong contributions to classification performance. The Accuracy diagram also indicates that Nickname, Fare, and Pclass perform well. Considering two diagrams, we should prioritize the following features: Nickname, Fare, Pclass, Sex, SibSp, and Age in the model.

```
c("Nickname", "Fare", "Sex", "Age", "Pclass", "SibSP")
```

Finally, in the Neural Networks code, I make the code itself to select best performance in features, the outcome show that the best features are Nickname and Fare.

The best feature set in Titanic NN model: Nickname + Fare

Model building:

GBM-XGBoost and Neural Networks input and output are similar to article model, so I will not describe it again.

GBM-XGBoost and Neural Networks Hyperparameters:

	Neural Networks	Neural Networks 10 K fold cross validation		XGBoost	XGBoost 10 K fold cross validation
Hidden Layer Size		1 to 10	Booster	gbtree	gbtree
Finded Best		2	objective	binary	binary
Weight Decay		0.01, 0.1	Learning Rate(eta)	0.1	0.1
Finded Best		0.01	MAX Depth	3	3
Max Iterations		100, 200	Number of Rounds	100	100
Finded Best		100	Subsample	0.8	0.8
Cross-Validation Method			colsample_bytree	0.8	0.8
Number		10	scale_pos_weight(Titanic data is imbalance)	1.6047	1.6047
			Cross-Validation		10 k fold

New method Result:

The result of predict Titanic passengers survival rate using different classifiers				
2 New method, GBM-XGBoost and Neural Networks				
	Accuracy %	Precision %	Recall %	F1-Score %
GBM-XGBoost	89.71	82.63	90.79	86.52
GBM-XGBoost 10 K fold cross validation	91.15	82.49	96.05	88.75
Neural Networks	96.65	91.57	1	95.60
Neural Networks 10 K fold cross validation	96.65	91.57	1	95.60

In Neural Networks, without or with cross validation the result is the same.

Compare original result with new method:

Of my individual opinion. Actually, I think it is hard/unnecessary to compare result between original and new method. They algorithm are not the same. First, each algorithm will use it logic to analysis Titanic data and print out confusion matrix, thereof my new method outcome must not the same with article. Second, even I use (my new method new features) in original 6 algorithm, it may not valid. Because each algorithm has its own unique understanding, the best features of algorithm A, may not equal to the best features of algorithm B.

So actually, we cannot compare in those algorithms in detail, the thinks we only can compare is which algorithm have the best performance. Of course, the except case is compare in XGBoost with Decision Tree and Random Forest. Since 3 model is tree-based model. In the performance metrics XGBoost is better than article Decision Tree and Random Forest. In XGBoost each new tree will try correct errors made by the previous tree, each tree will be learning from the residuals of the last one.

Compare performance and conclusions:

Without k-fold cross - validation				
	Accuracy	Precision	Recall	F1-Score
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GBM-XGBoost 10 K fold cross validation	91.15	82.49	96.05	88.75
Neural Networks	96.65	91.57	1	95.60
Neural Networks 10 K fold cross validation	96.65	91.57	1	95.60

Overall, of comparing original result with new method. XGBoost without cross validation is better than KNN, decision tree, Random Forest. Worse than Logistic, SVM, Naïve Bayes. XGBoost with 10 K fold CV is better than all article 10 k fold algorithm. Neural Networks without CV is better than all article algorithm except Naïve Bayes. Neural Networks with 10 K fold CV is better than article 10 k fold algorithm.

This comparison shows modern techniques like GBM-XGBoost and Neural Networks is significantly improve performance in classification tasks. If only see comparison in original result with new method, Neural Networks stand out as the best choice for this dataset.

In summary, the new methods perform better than the original models on the Titanic dataset because they have better flexible settings (hyperparameters) to manage complexity and can capture complex patterns in the data. Nonlinear relationships, interactions between features, and class imbalance can all affect model performance, and GBM-XGBoost and Neural Networks are better suited to handle these challenges than traditional models.